Experiment design

Experiment: An experiment is a well defined act on an investigation conducted to discover underlying facts about the phenomenon, which are utilized to test some hypothesis of interest. eg: (wheat with other varieties)

Treatment: A treatment is a procedure or method whose effect is to be measured and compare with others. ie. fertilizer dose.

Experimental unit: An experimental unit is a piece of experimental materials which a treatment is applied, a plot of land

Yield: The quantity which is measured from the Experimental unit.

Block: A group of experimental units which are internally homogenous.

Design of experiment:
The formation of a set of rules and principles according to which an experiment is to be conducted to collect the appropriate data and lead to valid inferences for the problem.

Purpose:
i) Main purpose collected the maximum amount of necessary information for the problem at a minimum cost (time and resources)
ii) Increase the accuracy of the result.

Basic Principle of experimental design

- **Replication**: Repetation of the basic treatment under investigation. Sometime in agriculture, replication leading to different soil and other conditions.
- **Randomisation**: The process of distributing the treatments purely by chance in such a way to the experimental unit, so that any experimental unit is equally likely to receive any treatment.
- **Local control**: Local control is the procedure to controlling and reducing error variation by arranging the experimental unit in Block.

ANOVA: Analysis of variance is the process of partitioning and decomposing total variation contained in data into various independent components.

Total variation can be classified as i) Assignable causes ii) Non assignable causes.

objectives:
1) It identify the causes of variation and sort out corresponding components of variation with associated degree of freedom
2) It provide for test of significance based on F distribution

assumption
- observation and errors are independently distributed.
- observation conform to normal distribution with equal variance.
- Treatment and environmental effects are additive.

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Basic Deg Design: i) CRD ii) RBD iii) LSD ~~iv) split plot~~

i) CRD - Completely randomised design.
In which the selected treatment are allocated or distributed to the experimental unit completely at random.
only two principle: 1) replication and 2) Randomisation.

Layout: The layout of design indicates the placement of treatments to the experimental plots according to the condition of the design.

$$y_{ij} = \mu + t_i + e_{ij}$$

advantage:
1) layout and analysis is easy
2) design is flexible
3) All measure are independent

disadvantages:
1) small number of treatment.
2) homogyous field no found in ag.
↳ seldom u.

ii) RBD: - Randomised Block Design
In which the whole set of experimental units are arranged in several block which are internally homgygous and externally heterogygous. Then the treatment are randomly allocated within each block, ADD Blocks as local control measure.
must popular involves all three basic principles (R, R, LC)
Block → Control the heterogeneity and variability among the experimental units.

analysis of RBD:
The linear model is, $y_{ij} = \mu + t_i + b_j + e_{ij}$ $\begin{pmatrix} i = 1,2,3 \ldots k \\ j = 1,2,3 \ldots n \end{pmatrix}$

where, y_{ij} → observation of ith treatment at jth block
μ → is the general mean effects.
t_i → effects due to ith treatment
b_j → effects due to jth block
e_{ij} → random error components.

let $y_{i.} = \sum_j y_{ij}$ → total of all observation for treatment i
$y_{.j} = \sum_i y_{ij}$ → total of all observation for block j.
$y_{..} = \sum_i \sum_j y_{ij}$ → Grand total of everything.
$\bar{y}_{i.} = \frac{y_{i.}}{r}$ → mean of treatment i
$\bar{y}_{.j} = \frac{y_{.j}}{k}$ → mean of block j
$\bar{y}_{..} = \frac{y_{..}}{rk}$ → the grand mean.

Getting the least square estimation to get γ, t_i and b_j

$$S = \sum\sum e_{ij}^2 = \sum\sum (y_{ij} - \gamma - t_i - b_j)^2$$

it gives three normal equations,

$$\hat{\gamma} = \bar{y}_{..} \rightarrow \text{grand mean}$$
$$\hat{t}_i = \bar{y}_{i.} - \bar{y}_{..} \rightarrow (\text{treatment mean} - \text{g m})$$
$$\hat{b}_j = \bar{y}_{.j} - \bar{y}_{..} \rightarrow (\text{Block mean} - \text{g m})$$

The total ss. can be partitioned into three components as follows

$$\sum_i^k \sum_j^n (y_{ij} - \bar{y}_{..})^2 = \sum_i^k \sum_j^n (y_{i.} - \bar{y}_{..})^2 + \sum_i \sum_j (y_{.j} - \bar{y}_{..})^2 + \sum_i \sum_j (y_{ij} - \bar{y}_{i.} - \bar{y}_{.j} + \bar{y}_{..})^2$$

$$= n \sum_i (y_{i.} - \bar{y}_{..})^2 + k \sum_j (y_{.j} - \bar{y}_{..})^2 + \sum_i \sum_j (y_{ij} + \bar{y}_{i.} + \bar{y}_{.j} + \bar{y}_{..})^2$$

Thus we have,

Total ss = Treatment ss + Block ss + Error ss.

Mainly here we want to test the hypothesis test here

i) H_0; $t_1 = t_2 = \dots = t_k$; all treatment are same

here the, $$F_1 = \frac{n \sum_i (y_{i.} - \bar{y}_{..})/(k-1)}{\sum_i \sum_j (y_{ij} - \bar{y}_{i.} - \bar{y}_{.j} + \bar{y}_{..})/(k-1)(n-1)}$$

ii) H_1; $b_1 = b_2 = \dots = b_n$; all block effects are same

$$F_2 = \frac{k \sum_j (y_{.j} - \bar{y}_{..})/(n-1)}{\sum_i \sum_j (y_{ij} - \bar{y}_{i.} - \bar{y}_{.j} + \bar{y}_{..})/(n-1)(k-1)}$$

Calculation for different sum of square

Total sum of square, $T_0 = \sum_i^k \sum_j^n (y_{ij} - \bar{y}_{..})^2 = \sum_i \sum_j y_{ij}^2 - CF$

Treatment sum of square, $T = \sum_i \sum_j (y_{i.} - \bar{y}_{..})^2 = \sum_i \sum_j y_{i.}^2/n - CF$

Block " " " , $B = \sum_i \sum_j (y_{.j} - \bar{y}_{..})^2 = \sum_i \sum_j y_{.j}^2/k - CF$

here, CF = correction factor $= \frac{y_{..}^2}{nk}$

Error SS, $E = T_0 - T - B$

Now ANOVA Table:

Source of variation	df	SS	M.S	F_{cal}
Treatment	$(K-1)$	T_o	$T' = T/K-1$	$F_1 = \frac{T'}{E'}$
Block	$(n-1)$	B	$B' = B/n-1$	
Error	$(k-1)(n-1)$	E	$E' = E/(k-1)(n-1)$	$F_2 = \frac{B'}{E'}$
Total	$nk-1$	T_o	-	

If the calculated values of F_1 with $(k-1)$ and $(k-1)(n-1)$ df is greater then tabulated values then Null hypothesis can be rejected on otherwise accept.

Same for F_2

LSD – Latin square Design

LSD is a design in which experiment unts are annranged in a complete bocks in two different ways, called row and column. Then the selected treatment are randomly allocated in a way so that all row and column are consists of all the treatments once. So this why square.

linear model, $y_{iijs} = \mu + r_i + c_j + t_s + e_{iijs}$

y_{iijs} → The observation of the sth treatment in ith row and jth column
μ → is the general mean
r_i → effects due to ith row
c_j → " " " jth column
t_s → effect due to sth treatment.
e_{iijs} → error which normally distributed.

Comparison among basics design:

Features	CRD	RBD	LSD
Setting of experiments	simple and easy	easy but require to set up block of appropriate size	more difficult as row + colm.
No of treatment	No limitations	if too many difficult reduced → blm	5 - 10
Degree of freedom for error	Max		row → row + colm
Control of variability	No scope	control by block	two ways (c + r)
Missing value problem	No difficulty	Not difficult but effect less	problem → several loss
Analytical problem	Minor problem	complicated	may badly upset.
Test of unequal error variance	simple	difficult	very difficult

- split-plot design, advantages
- other topic → The other questions (from pdf → claude)

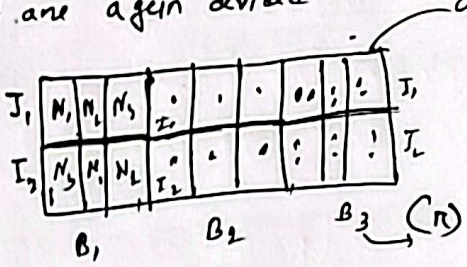
Split-plot design

→ whole plot / main plot

When there are two factors, and one factor require large plot (Irrigation, tillage, date of sowing) and second in small plot then this design used. Here the Error will be higher than factorial RBD ! → sub plot

Layout: 2 Irrigation (I_1, I_2) 3 Nitrogen (N_1, N_2, N_3)

(i) Divide block (ii) each block divide into whole plot (iii) Each plot are again devided



Breeder: whole plot (Nitrogen) sub plot (varieties)

Agronomy: " (varieties) sub plot (Nitrogen)

because sub plot precision higher (B)

better factorial RBD

Anova table

Source of Variation	df	SS	MS	Fcal
1) replication (R)	(R-1)			$A \rightarrow \frac{MS_a}{MS_{Er\,a}}$
2) Main plot factor (a)	(a-1)			
3) Error (a)	(R-1)(a-1)			$B = \frac{MS_b}{MS_{Er\,b}}$
4) sub plot factor (b)	(b-1)			$A \times B = \frac{MS_{(a \times b)}}{MS_{Er\,b}}$
5) Interaction (a×b)	(a-1)(b-1)			
6) Error (b)	(Total - a(R-1))			a(R-1)(b-1)
7) Total	abR - 1			

Factorial RBD

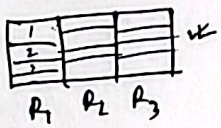
When two or more factor are in the experiment.

N dose: 50, 100 ; K dose: 20, 60

$T_1 \rightarrow 50+20$ $T_2 \rightarrow 50+60$ $T_3 \rightarrow 100+20$ $T_4 \rightarrow 100+60$

Layout : (i) Block (ii) Each block contain randomized treatment.

↳ No of replication



Anova

Source of Variation	df RBD	df fac RBD	SS	MS	F
Replication	r-1	(r-1)			
Treatment	t-1	(ab-1)			
Nitrogen (a)		(a-1)			
phosphorus (b)		(b-1)			
N×P (a×b)		(a-1)(b-1)			
Error	(r-1)(t-1)	(r-1)(ab-1)			
Total	rt-1	abr - 1			

here t = N×P

t = a×b